

AGRIM JAIN

[linkedin.com/in/agrim-jain](https://www.linkedin.com/in/agrim-jain) ◇ agrimportfolio.netlify.app

Meerut, Uttar Pradesh, India

+91 9760939593 ◇ agrimjain015@gmail.com ◇ github.com/AgrimJain-quantum

EDUCATION

Bachelor of Technology in Electrical and Electronics Engineering Manipal University Jaipur Current CGPA : 5.41	2023 – Present
---	----------------

SKILLS AND INTERESTS

Programming Languages	Python, C, SQL
Machine Learning	Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn
MLOps & Tools	Git, GitHub, Streamlit, Flask, CI/CD Basics
Core Interests	Machine Learning, AI Systems, MLOps, Energy Systems, Load Forecasting, Research-driven Development
Creative Skills	Video Editing, Photography

PROJECTS

Short-Term Electricity Load Forecasting (Ongoing) <i>Research Project</i>	2026 – Present <i>Python, Scikit-Learn, XGBoost, LightGBM, LSTM, Optuna</i>
---	--

- Conducting a comparative study of Machine Learning and Deep Learning models for short-term electricity demand forecasting.
- Implementing XGBoost, LightGBM, Random Forest, and LSTM models using real-world electricity consumption datasets.
- Performing feature engineering, time-series preprocessing, and hyperparameter optimization using Optuna.
- Evaluating forecasting performance using RMSE, MAE, and MAPE to identify the most effective modeling approaches.
- Investigating the effectiveness of optimized classical ML methods compared to deep learning architectures on structured energy datasets.

Multi-Dataset Power Forecasting System (Ongoing) <i>Research Project</i>	2026 – Present <i>Python, TensorFlow, XGBoost, SciPy</i>
--	---

- Developing a forecasting framework to benchmark LSTM, GRU, XGBoost, and Random Forest models across multiple power grid datasets.
- Conducting cross-dataset evaluation using Delhi electricity demand data and UK National Grid demand data.
- Building reproducible experimentation pipelines for model comparison and forecasting research.
- Analyzing model robustness, scalability, and generalization across geographically distinct energy systems.
- Exploring deployment considerations for practical energy forecasting applications.

Audio Feature Music Recommendation Engine <i>Machine Learning Project</i>	2025 <i>Python, Scikit-Learn, Streamlit</i>
---	--

- Built a content-based music recommendation system using Spotify-style audio features.
- Applied feature normalization using StandardScaler and similarity matching using Cosine Similarity.
- Developed an interactive Streamlit application for generating personalized music recommendations.
- Performed feature engineering and similarity-based ranking to improve recommendation quality.

RESEARCH EXPERIENCE

-
- Currently conducting independent research in electricity load forecasting and power demand prediction.

- Investigating the comparative performance of Machine Learning models (XGBoost, LightGBM, Random Forest) and Deep Learning models (LSTM, GRU) for energy forecasting applications.
- Exploring forecasting model generalization across geographically distinct electricity grids through cross-dataset experimentation.
- Performing systematic evaluation using industry-standard forecasting metrics and reproducible research methodologies.
- Preparing ongoing research outcomes for future publication and advanced academic work in AI-driven energy systems.

POSITION OF RESPONSIBILITY

Joint Head of Media, IEEE CS MUJ

2025

IEEE Computer Society, Manipal University Jaipur

- Managed media operations and creative content for IEEE CS events and activities.
- Worked on video editing, promotional design, and event coverage.
- Coordinated with student teams for technical and media initiatives.

EXTRA-CURRICULAR

- Active in photography and video editing projects.
- Interested in AI research, intelligent systems, and technology-driven innovation.
- Regularly participate in self-driven technical learning and project building.